ELEN E6885: Final Project Instruction

This course project aims to provide hands-on experience and an opportunity of research in the field of reinforcement learning. Based on your own background, you are free to select any RL-related topic for the project. To inspire ideas, you can consider RL publications in NeurIPS, ICML, RLDM, AAMAS, etc.

I. IMPORTANT INFORMATION

1) Final report submission: Dec. 16th
2) We recommend a team of three to five members. It is fine for only one or two members in a team. More team members are expected to deliver more results and novelty.
3) Grading policy: a. Contribution (20%); b. Novelty (20%); c. Writing (20%) ; d. Completion (40%)

II. PROJECT FINAL REPORT

The final report should be no more than 5 pages excluding references and the statement of team member contributions. If needed, the theoretical proofs or the detailed methodology of implementing an algorithm can be submitted as supplementary materials. All necessary information needs to be submitted such that all results in the report can be recovered (for grading). The report should following the structure as follows:

1) Title, Authors
2) Abstract
3) Introduction
4) Background/Related Work and your novel contribution
5) Approach: algorithm development/theoretic results
6) Experiment results
7) Conclusion
8) References
9) A statement on the contributions of each member of the team.

The report needs to be written in the double-column ICML format. Only **ONE** member in your team needs to submit the report and the codes (uploaded to GitHub). Exceptional project results will be further reviewed by the instructor and other professors, and then recommended to submit to a machine learning conference.

III. SUGGESTED PROJECT TOPICS

(you can always choose your own interested topics)

1. Smart Energy Management: Design a system that uses reinforcement learning to optimize energy consumption in a smart home, adjusting usage based on patterns and preferences.
2. Stock Trading Agent: Develop an agent that learns to make buy/sell decisions based on historical stock market data, optimizing for maximum return.
3. Personalized Learning System: Create a reinforcement learning-based educational tool that adapts to individual student learning styles and paces.
4. Traffic Signal Optimization: Simulate a traffic management system where signals learn to optimize traffic flow at intersections using reinforcement learning.